Topic 13B: Entropy
12. understand that, since some endothermic reactions can occur at room temperature, enthalpy changes alone do not control whether reactions occur
13. know that entropy is a measure of the disorder of a system and that the natural direction of change is increasing total entropy (positive entropy change)
14. understand why entropy changes occur during: i changes of state ii dissolving of a solid ionic lattice iii reactions in which there is a change in the number of moles from reactants to products *Students should be able to discuss typical reactions in terms of disorder and enthalpy change, including:* o *dissolving ammonium nitrate crystals in water* o *reacting ethanoic acid with ammonium carbonate* o *burning magnesium ribbon in air* o *mixing solid barium hydroxide, $Ba(OH)_2.8H_2O$, with solid ammonium chloride.*
15. understand that the total entropy change in any reaction is the entropy change in the system added to the entropy change in the surroundings, shown by the expression: $\Delta S_{total} = \Delta S_{system} + \Delta S_{surroundings}$
16. be able to calculate the entropy change for the system, ΔS_{system} , in a reaction, given the entropies of the reactants and products
17. be able to calculate the entropy change in the surroundings, and hence ΔS_{total} , using the expression: $\Delta S_{surroundings} = -\dfrac{\Delta H}{T}$
18. know that the balance between the entropy change and the enthalpy change determines the feasibility of a reaction and is represented by the equation $\Delta G = \Delta H - T\Delta S_{system}$
19. be able to use the equation $\Delta G = \Delta H - T\Delta S_{system}$ to: i predict whether a reaction is feasible ii determine the temperature at which a reaction is feasible
20. be able to use the equation $\Delta G = -RT \ln K$ to show that reactions which are feasible in terms of ΔG have large values for the equilibrium constant and vice versa
21. understand why a reaction for which the ΔG value is negative may not occur in practice